23. Prove that given a real number x there exist unique numbers n and ε such that $x = n - \varepsilon$, n is an integer, and $0 \le \varepsilon < 1$.

$\boxed{\text{Solution:}}$ We know, that for any arbitrary real number x, there exists an integer n such that $x \le n < x+1$. ??

We shall prove that such an integer n is unique.

~~Let ∃ an integer n_1 s.t. $x \le n_1 < x+1$ ... $(x+1)-(x+1) \Rightarrow 0 < n - n_1 < 0$. This can only happen~~ [illegible]

$\because x \le n \;\therefore (x+1) \le n+1 \quad \because n < x+1 \;\therefore (n-1) < x.$

The successor and predecessor of n all lie outside the range $[x, x+1)$.

$\therefore$ The rest of the integers must lie outside of this range.

$\boxed{\text{So, } n \text{ is unique}}$ $\square$. $\boxed{\text{Note:}}$ The argument shown in Problem 22 for the uniqueness of n is Wrong. That argument will be similar to the argument presented here.

Let, $x = n - \varepsilon$. $\therefore n = x + \varepsilon$ $\quad x \le n \Rightarrow x \le x + \varepsilon \Rightarrow \boxed{0 \le \varepsilon}$

$n < x+1 \Rightarrow x + \varepsilon < x+1 \Rightarrow \boxed{\varepsilon < 1}$. $\boxed{\therefore 0 \le \varepsilon < 1}$

$\therefore$ There exists a real no. ε, s.t. $x = n - \varepsilon$ and $0 \le \varepsilon < 1$. We have to prove that such an ε is unique.

Let, $x = n - \varepsilon_1$. It's easy to show that $0 \le \varepsilon_1 < 1$.

$\therefore n - \varepsilon = n - \varepsilon_1 \Rightarrow -\varepsilon = -\varepsilon_1 \Rightarrow \boxed{\varepsilon = \varepsilon_1}$. $\square$

24. Use forward reasoning to show that if x is a nonzero real number, then $x^2 + \frac{1}{x^2} \ge 2$.

$\boxed{\text{Solution:}}$ We know that for all non-zero real numbers x, $\left(x - \frac{1}{x}\right)^2 \ge 0$.

$\therefore x^2 + \frac{1}{x^2} - 2 \cdot x \cdot \frac{1}{x} \ge 0^2 \Rightarrow x^2 + \frac{1}{x^2} - 2 \ge 0 \Rightarrow \boxed{x^2 + \frac{1}{x^2} \ge 2}$. $\square$